Lecture 19 (*April 5*)

Reading: Section 5.1
Viewing: Lecture 18

Consider any linear transformation $\phi : \mathbb{R}^n \to \mathbb{R}^n$, represented by an $n \times n$ square matrix A.

Definition 17. *The* ***determinant*** *of the* <u>*square*</u> *matrix A is the number:*

$$\boxed{\det A} \tag{149}$$

defined as the factor by which the linear transformation ϕ scales the n–dimensional volumes of regions in $\mathbb{R}^n$ (the determinant can be negative if ϕ switches left-handedness to right-handedness).

Let's take for instance the linear transformations of $\mathbb{R}^2$ considered immediately after Fact 11. Two-dimensional volume just means area. Any reflection preserves the magnitude of areas but switches handedness (the mirror images of left hands are right hands), hence:

$$\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1 \tag{150}$$

Projections flatten regions $R \subset \mathbb{R}^2$ to line segments, which have area 0, so therefore:

$$\det \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = 0 \tag{151}$$

Scaling transformations increase areas by the product of the scaling factors, and therefore:

$$\det \begin{bmatrix} \lambda & 0 \\ 0 & \mu \end{bmatrix} = \lambda\mu \tag{152}$$

Finally, rotations and shearings preserve areas, so we have:

$$\det \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = 1$$

$$\det \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} = 1 \tag{153}$$

In general, the determinant of a 2×2 matrix is given by the following formula:

$$\boxed{\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc} \tag{154}$$

and indeed, the determinant plays an important role in the formula for the inverse of a 2×2 matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \tag{155}$$

But before we work out general means for computing determinants, let us spell out two key facts.

Fact 13. *For any $n \times n$ matrices A and B, we have:*

$$\det AB = (\det A)(\det B) \tag{156}$$

Let's give a quick proof of this fact: if A and B correspond to linear transformations ϕ and ψ, respectively, then we have seen in (135) that AB corresponds to the composition $\phi \circ \psi$ of the two transformations. Then (156) is simply saying that the rate at which $\phi \circ \psi$ scales volumes is equal to the product of rates at which ϕ and ψ individually scale volumes. Indeed, if you take any region $R \subset \mathbb{R}^n$ and assume that:

$$\begin{cases} \psi \text{ takes } R \text{ to some region } R' \\ \phi \text{ takes } R' \text{ to some region } R'' \end{cases} \quad \Rightarrow \quad \phi \circ \psi \text{ takes } R \text{ to the region } R''$$

With this in mind, we have:

$$\text{LHS of (156)} = \frac{\text{vol } R''}{\text{vol } R} = \frac{\text{vol } R''}{\text{vol } R'} \cdot \frac{\text{vol } R'}{\text{vol } R} = \text{RHS of (156)}$$

which proves (156). Caveat: it is not true that $\det(A + B) = \det A + \det B$, in general!

Fact 14. *Row operations have the following effect on determinants:*

- *adding a multiple of one row to another leaves the determinant unchanged*
- *exchanging two rows multiplies the determinant by -1*
- *multiplying a single row by λ has the effect of multiplying the determinant by λ*

The analogous properties still hold if the word "row" is replaced by "column" everywhere.

Fact 14 actually follows from Fact 13, as we will now proceed to show. Recall from Lecture 2 that performing the three kinds of row operations above on a matrix A is precisely achieved by multiplying A on the left by elimination/permutation/diagonal matrices. Then the statements in the bullets of Fact 14 become equivalent to:

$\det E_{ij}^{(\lambda)} A = \det A$	which happens because	$\det E_{ij}^{(\lambda)} = 1$
$\det P_{ij} A = -\det A$	which happens because	$\det P_{ij} = -1$
$\det D_i^{(\lambda)} A = \lambda \cdot \det A$	which happens because	$\det D_i^{(\lambda)} = \lambda$

The fact that P_{ij}, $D_i^{(\lambda)}$, $E_{ij}^{(\lambda)}$ have determinants -1, λ, 1 respectively, is simply a natural generalization of (150), (152), (153). Fact 14 has the following important consequence:

$$\textbf{If } A \textbf{ is a square matrix, then } \boxed{\det A = \pm \text{ product of pivots of REF}(A)} \tag{157}$$

which means that a computationally efficient way to compute the determinant of a matrix is to just put it in row echelon form and multiply its pivots. Indeed, Fact 14 implies that:

$$\det A = \pm \det U \tag{158}$$

where U is the row echelon form of A, and the sign $\pm$ is simply (-1) raised to the number of row exchanges required by Gaussian elimination. A square matrix in row echelon form is the same thing as upper triangular, and its pivots are just its diagonal entries. Therefore, to get from (158) to (157) we need to invoke the following fact, which simplifies a lot of computations:

the determinant of a triangular/diagonal matrix is the product of its diagonal entries

The fact above is true for both upper and lower triangular matrices, as well as for diagonal ones:

$$\det \begin{bmatrix} d_1 & 0 & \dots & 0 \\ * & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \dots & d_n \end{bmatrix} = \det \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} = \det \begin{bmatrix} d_1 & * & \dots & * \\ 0 & d_2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} = d_1 d_2 \dots d_n \tag{159}$$

where the $*$'s denote arbitrary numbers. Let's prove (159) in the case of an upper triangular matrix (the other two cases featured in the formula above are analogous). We have:

$$\begin{bmatrix} d_1 & * & \dots & * \\ 0 & d_2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} = D_1^{(d_1)} \dots D_n^{(d_n)} \begin{bmatrix} 1 & \# & \dots & \# \\ 0 & 1 & \dots & \# \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} \tag{160}$$

where the $*$'s and $\#$'s denote arbitrary numbers. By Gauss-Jordan elimination, the right-most matrix in (160) can be reduced to the identity matrix I_n using only the first bullet in Fact 14. As:

$$\det I_n = 1$$

(the identity linear transformation does not change volumes) then applying Fact 13 to (160) yields:

$$\det \begin{bmatrix} d_1 & * & \dots & * \\ 0 & d_2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} = \det \left(D_1^{(d_1)} \dots D_n^{(d_n)} \right) = \left(\det D_1^{(d_1)} \right) \dots \left(\det D_n^{(d_n)} \right) = d_1 \dots d_n$$

as we needed to show in (159). This completes the proof of (157).

Remark. *Another way to restate formula* (157) *is to remember that any square matrix A has a factorization of the following form for a suitable permutation matrix P:*

$$PA = LDU$$

where L and U are lower/upper triangular with all 1*'s on the diagonal, and D is a diagonal matrix. Applying determinant and Fact 13 to the formula above gives us;*

$$(\det P)(\det A) = (\det L)(\det D)(\det U)$$

and then (157) *follows from the fact that* $\det P = (-1)$ *raised to the number of row exchanges, while* $\det L = \det U = 1$ *and* $\det D =$ *product of pivots.*

If a square matrix A has a full row of zeroes (or a full column of zeroes, for that matter), then its determinant is 0: this is simply a consequence of (157) and the fact that one of the pivots would be 0 if we had a full row of zeroes. But we also have the following more general property:

$$\textbf{If the rows/columns of } A \textbf{ are linearly dependent, then } \det A = 0 \tag{161}$$

Indeed, if the rows were dependent, then one of them (say row i) would be a linear combination of the other ones. But then by successively subtracting multiples of the other rows from row i, which we know does not change the determinant, we would arrive at a matrix with a full row of zeroes. Since such a matrix would have a zero pivot, it has determinant 0, thus establishing (161).

Expanding on the discussion above, we know exactly when a matrix has linearly dependent rows or columns: precisely when it is singular, i.e. not invertible. Therefore, (161) can be restated as:

Fact 15. *A square matrix A is non-singular (invertible) if and only if* $\det A \neq 0$.

Indeed, if the matrix A had an inverse A^{-1}, then applying determinant and Fact 13 to the formula $AA^{-1} = I$ would imply that:

$$(\det A)(\det A^{-1}) = \det I = 1 \quad \Rightarrow \quad \det A^{-1} = \frac{1}{\det A} \tag{162}$$

This immediately implies that the determinant of an invertible matrix cannot be 0. Finally, we give one more fact for the road:

Fact 16. *For any square matrix A, we have* $\boxed{\det A = \det A^T}$.

If Q is an orthogonal matrix (see Definition 15), i.e. $Q^T = Q^{-1}$, then (162) and Fact 16 imply that its determinant can only be 1 or -1.